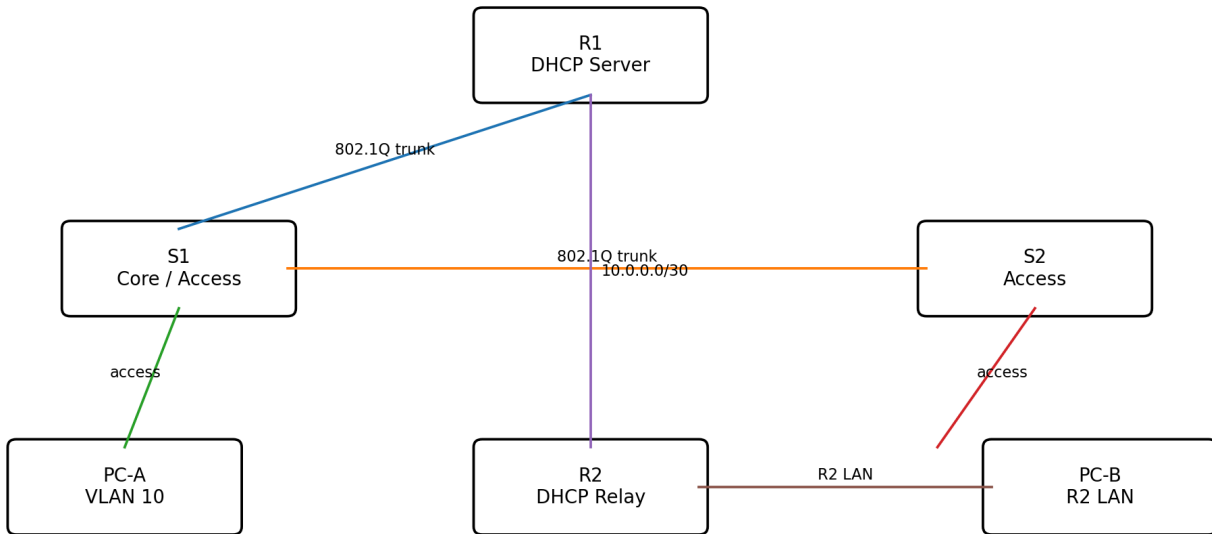


CCNA Intermediate Practice Exam — Draft

IPv4 only • VLANs • 802.1Q Trunking • DHCPv4 • DHCP Relay • Switch Security

CCNA Intermediate Practice Exam — IPv4 / VLAN / DHCP / Switch Security



Goal: configure the network from a blank topology and verify end-to-end IPv4 connectivity.

Instructions

- Start with blank/default devices. Do not use a prepared configuration.
- Use IPv4 only. IPv6 is not part of this test.
- You may use show commands and the ? command to verify syntax.
- Configure the network so that the final result has end-to-end IPv4 connectivity.
- Do not ask for the solution while testing yourself; record your commands and verification output separately.

1. Addressing & VLAN Plan

Use the base network 192.168.50.0/24 and subnet it yourself. Choose appropriate IPv4 subnets for the requirements below.

Network	Purpose	Host requirement	Network/CIDR	Gateway	Notes
VLAN 10	Clients	50 hosts	192.168.1.0-63	192.168.1.1	PC-A lives here
VLAN 20	Management	25 hosts	192.168.1.64-95	192.168.1.65	S1/S2 management
R2 LAN	Remote clients	12 hosts	192.168.1.96-111	192.168.1.97	PC-B lives here
R1 ↔ R2	Transit	2 hosts	192.168.1.112-		Point-to-point

Record your final addressing plan here:

Device	Interface	IPv4 address	Mask	Default gateway
R1	G0/0/0			N/A
R1	G0/0/1.10			
R1	G0/0/1.20			
R1	G0/0/1.999	No IP		N/A
R2	G0/0/0			N/A
R2	G0/0/1			
S1	VLAN 20			

S2	VLAN 20			
PC-A	NIC	DHCP	DHCP	DHCP
PC-B	NIC	DHCP	DHCP	DHCP

2. Basic Device Configuration

1. Configure hostnames R1, R2, S1 and S2.
2. Disable DNS lookup on all Cisco devices.
3. Configure an encrypted privileged EXEC password.
4. Configure console and VTY access with authentication.
5. Encrypt plaintext passwords.
6. Configure an appropriate warning banner.
7. Set the device clock.
8. Save running-config to startup-config.
9. Configure interface descriptions on all used links.

3. VLAN Configuration

Create the following VLANs on S1 and S2:

VLAN ID	Name	Purpose
10	Clients	User devices
20	Management	Switch management
999	Parking Lot	Unused ports
1000	Native	Native VLAN for trunks

10. Assign PC-A's switchport to VLAN 10 as a static access port.
11. Configure S1 and S2 management SVIs on VLAN 20 using the addresses you calculated.
12. Configure the correct default gateway on both switches.
13. Move all unused switchports to VLAN 999, make them static access ports, and administratively shut them down.
14. Verify the VLAN database and port assignments.

4. Inter-VLAN Routing & Trunking

15. Configure R1 G0/0/1 as the router-on-a-stick link to S1.
16. Create subinterfaces for VLAN 10 and VLAN 20 using 802.1Q encapsulation.
17. Configure VLAN 999 as the native VLAN on the R1 trunk and do not assign an IP address to the native subinterface.
18. Configure S1 ↔ S2 as a manually configured 802.1Q trunk.
19. Use VLAN 1000 as the native VLAN on the S1 ↔ S2 trunk.
20. Allow only VLANs 10, 20 and 1000 on the S1 ↔ S2 trunk.
21. Disable DTP on manually configured trunks.
22. Verify trunk status, native VLAN and allowed VLANs.
23. Test whether PC-A can reach its default gateway.

5. DHCPv4 on R1

24. Configure R1 as a DHCPv4 server for VLAN 10.
25. Configure a second DHCPv4 pool for the R2 LAN.
26. Exclude the first five usable addresses from each DHCP pool.
27. Configure the correct network and default gateway for each pool.
28. Set the domain name to ccna-practice.local.
29. Set the lease time to 2 days 12 hours 30 minutes.
30. Verify the DHCP pools, bindings and server statistics.
31. Renew PC-A's DHCP lease and record the assigned IPv4 address.

6. R2 Static Routing & DHCP Relay

32. Configure the R1 ↔ R2 transit link using your calculated /30 subnet.
33. Configure R2's LAN interface using the first usable address of the R2 LAN subnet.
34. Configure a static/default route so both routers can reach the remote LAN.
35. Configure DHCP relay on R2's LAN interface using R1's transit IP as the helper address.
36. Renew PC-B's DHCP lease.
37. Verify the DHCP binding on R1.
38. Verify that PC-B can ping its local gateway and R1.

7. Switch Security

39. Enable port security on the PC-A access port on S1.
40. Set the maximum number of secure MAC addresses to 2.
41. Use sticky MAC learning.
42. Use violation mode restrict.
43. Configure an aging time of 60 minutes.
44. Use inactivity aging.
45. On S2, configure the PC-facing access port with port security and a maximum of 1 secure MAC address.
46. Use sticky learning on S2 and violation mode protect.
47. Verify port-security status and the secure MAC address table.

8. DHCP Snooping, PortFast & BPDU Guard

48. Enable DHCP snooping globally on S2.
49. Enable DHCP snooping for VLAN 10.
50. Configure the uplink toward the DHCP server/network as trusted.
51. Rate-limit DHCP packets on the untrusted PC-facing port to 5 packets per second.
52. Verify DHCP snooping status.
53. Verify the DHCP snooping binding table after renewing the PC's address.
54. Enable PortFast on active PC-facing access ports.
55. Enable BPDU Guard on those PC-facing ports.
56. Verify PortFast and BPDU Guard using appropriate show commands.

9. Verification Challenge

Do not consider the exam complete until all relevant checks below succeed.

- show ip interface brief on R1 and R2
- show vlan brief on S1 and S2
- show interfaces trunk on S1 and S2
- show ip dhcp pool on R1
- show ip dhcp binding on R1
- show ip dhcp snooping on S2
- show ip dhcp snooping binding on S2
- show port-security interface <PC-A-port>
- show port-security interface <PC-B-port>
- show spanning-tree interface <PC-port> detail
- Ping from PC-A to its gateway, R2, PC-B and the opposite switch SVI where routing permits.
- Ping from PC-B to its gateway, R1 and PC-A.

10. Troubleshooting Scenarios

During the exam, deliberately test your ability to diagnose these situations. Do not change anything unless you can explain why.

57. PC-A receives an APIPA address instead of a DHCP address. Identify the likely Layer 2/Layer 3/DHCP causes and verify them.
58. The S1 ↔ S2 trunk is up, but VLAN 10 hosts on different switches cannot communicate. Identify what you would inspect.
59. DHCP works on VLAN 10 but PC-B never receives an address. Identify the required DHCP relay path.
60. A PC-facing port becomes err-disabled after a BPDU is received. Explain which security feature caused it and how you would verify it.
61. A switch's management SVI is down/down. Identify the Layer 2 condition required for the SVI to become operational.

11. Final Exam Checklist

- All devices have correct hostnames and basic security.
- All required VLANs exist on both switches.
- Unused ports are in VLAN 999 and shut down.
- Management SVIs are configured and reachable.
- Both trunks use the required native VLANs.
- DTP is disabled where manually required.
- R1 performs router-on-a-stick routing.
- R1 provides DHCPv4 for both DHCP networks.
- R2 relays DHCP requests to R1.
- Port security is active with the required settings.
- DHCP snooping is active with trusted/untrusted ports correctly configured.
- PortFast and BPDU Guard are enabled on PC-facing access ports.
- All required pings succeed.
- Configurations are saved.

12. Your Notes / Commands

[illegible]

